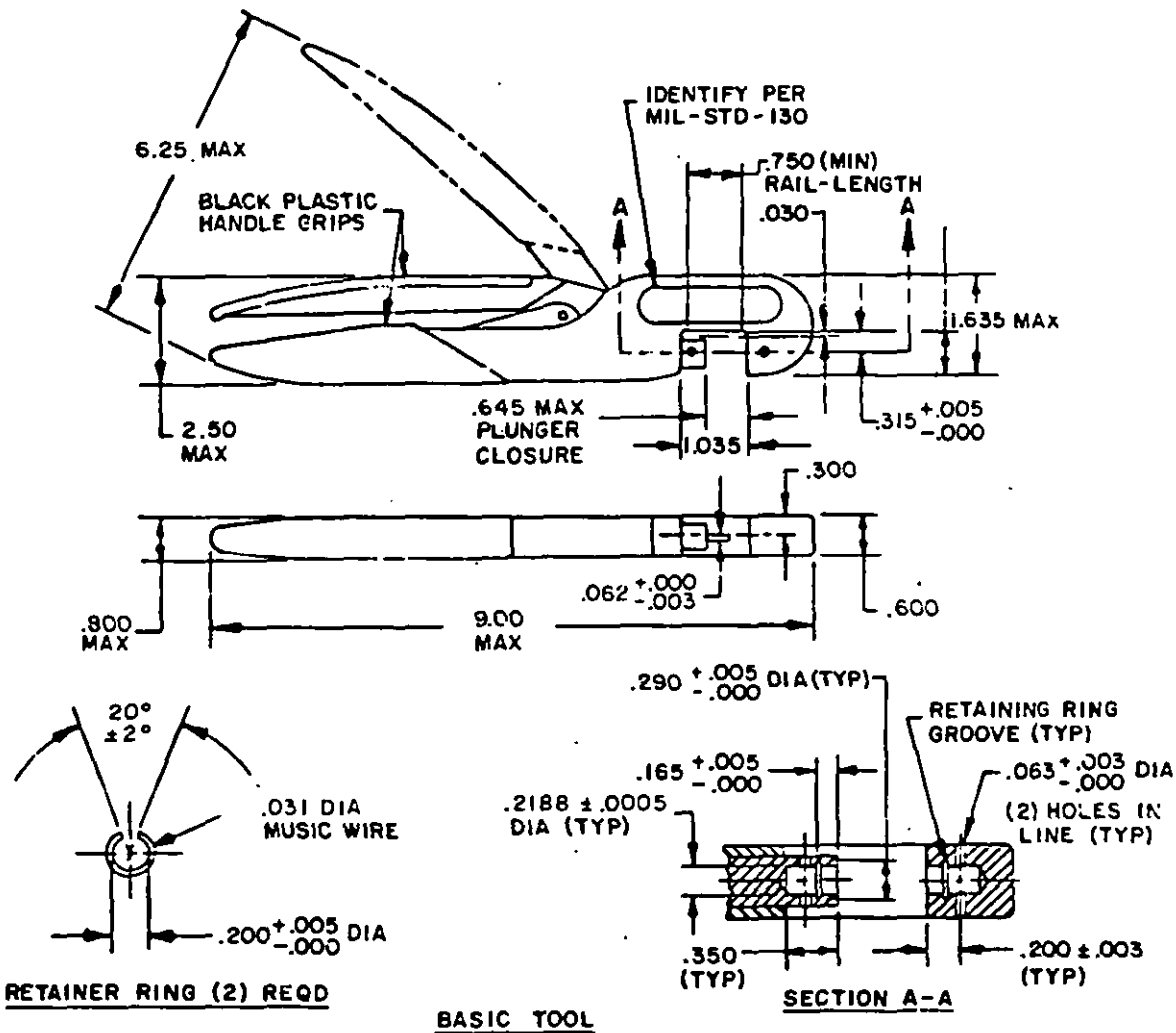
MILITARY SPECIFICATION SHEET

CRIMPING TOOLS, TERMINAL, HAND, WIRE TERMINATION, SMALL, FOR COAXIAL, SHIELDED CONTACTS, FERRULES, TERMINAL LUGS, SPLICES, AND END CAPS

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the crimping tools described herein shall consist of this specification sheet and the latest issue of MIL-C-22520.



BASIC TOOL

PART NO.	BODY COLOR
M22520/10-01	GOLD

FIGURE 1. Crimping tool.

AMSC N/A
FSC 5120

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil - Downloaded: 2026-01-19T12:46Z
Check the source to verify that this is the current version before use.

INCHES	MM	INCHES	MM
.000	0.00	.300	7.62
.0005	0.013	.315	8.00
.003	0.08	.350	8.89
.005	0.13	.600	15.24
.030	0.76	.645	16.38
.031	0.79	.750	19.05
.062	1.57	.800	20.32
.063	1.60	1.035	26.29
.165	4.19	1.635	41.53
.200	5.08	2.50	63.5
.2188	5.558	6.25	158.8
.290	7.37	9.00	228.6

NOTES:
1. Outline configuration optional within maximum dimensions shown.
2. For applicable dies see figures 2, 3, 4, and 5, and tables I, II, and III.
3. For inspection gauges see tables I, II and III, and figures 4 and 5.
4. Dies are interchangeable in the basic tool, when permanent die assembly is required, they may be pinned in place by means of MS171526 spring pins.
5. Dimensions are in inches.
6. Metric equivalents are given for general information only.
7. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals, for angles ±.50°.
8. The following information is supplemental to Compression force and Type Z tools paragraphs of MIL-C-22520.

Die No.	Test cable	Test ferrule ±.002 tol.	Compression force lbs. max.
M22520/10-03	RG-62/U	.261 I.D.,..297 O.D.	100

FIGURE 1 Crimping tool - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:46Z
Check the source to verify that this is the current version before use.

REQUIREMENTS:

Basic tool: See figure 1.

Weight: The weight of the basic tool shall not exceed 1.5 pounds.

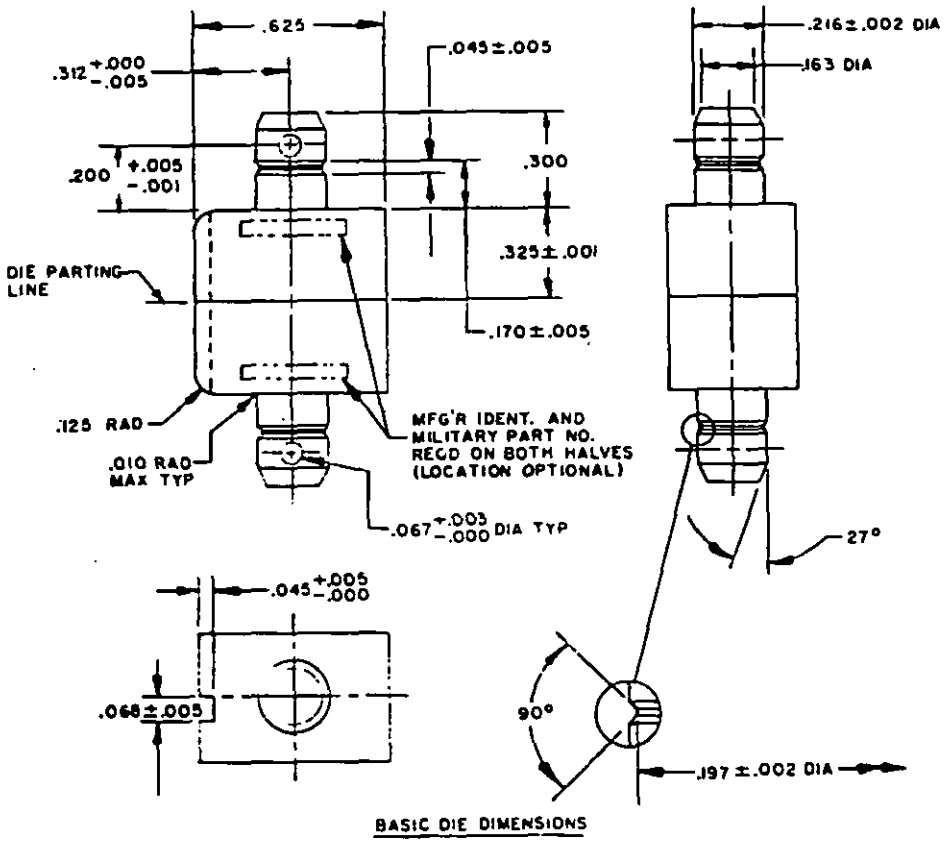
Rachet release force: Rachet release force on the handle shall be set at 30 to 50 pounds when tool is closed on a .650 ±.0005-inch (16.510 ±0.013 mm) die or gauge block.

Material: Dies shall be steel per QQ-S-628, class B, or SAE 0-6. Hardness C52-58.

Finish: Black oxide per MIL-C-13924.

Surface texture: Surface texture on crimping surfaces shall be 32 microinches in accordance with ASA B46.1-1962 (ANSI).

Source: https://assist.dla.mil -- Downloaded: 202640-19T12:46Z
Check the source to verify that this is the current version before use.



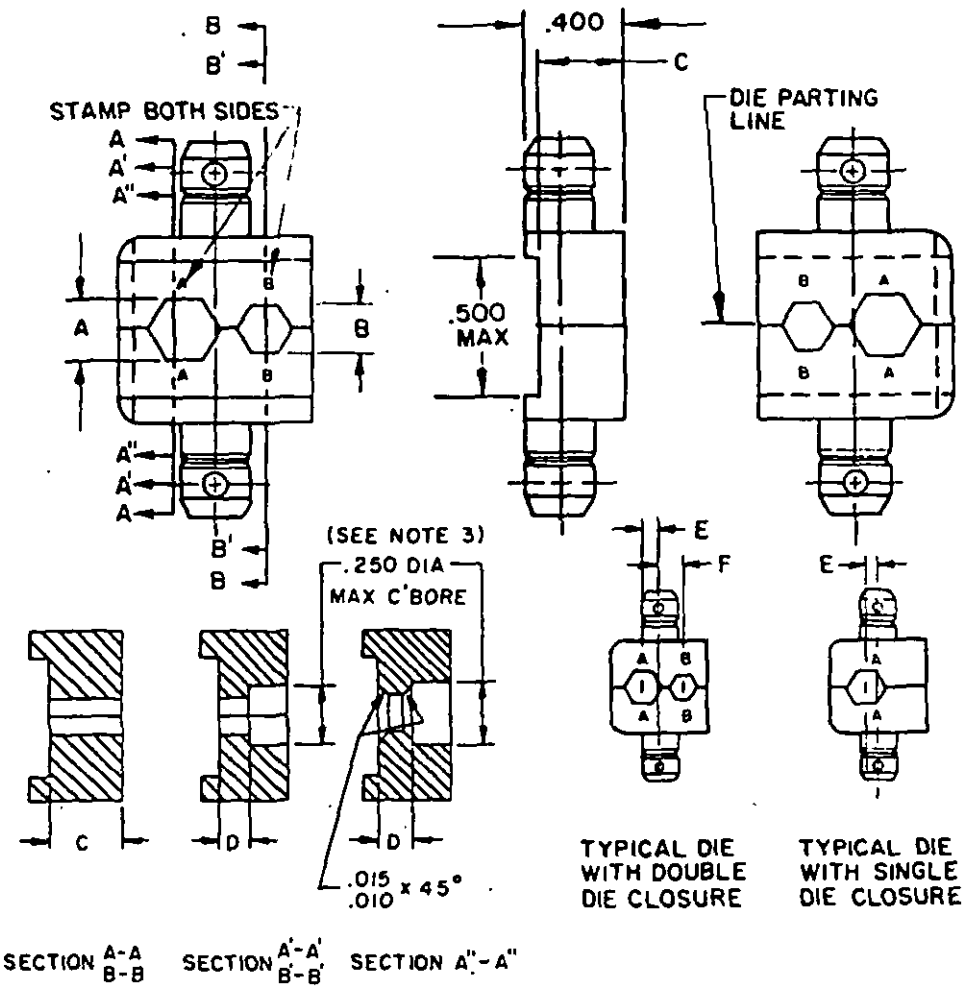
BASIC DIE DIMENSIONS

INCHES	MM
.000	0.00
.001	0.03
.002	0.05
.003	0.08
.005	0 13
.010	0.25
.045	1.14
.067	1.70
.068	1.73
.125	3.18
.163	4.14
.170	4.32
.197	5.00
.200	5.08
.216	5.49
.300	7.62
.312	7.92
.325	8.26
.625	15.88

NOTES:
1. Dimensions and requirements are applicable to both die halves.
2. For crimp die closure details see figures 3, 4 and 5, and table III.
3. Dimensions are in inches.
4. Metric equivalents are given for general information only.
5. Unless otherwise specified, tolerances are ±.010 (0.25 mm) for three place decimals, for angles ±.50°.
6. Remove all burrs. Break all edges and corners.

FIGURE 2. Basic die.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:46Z
Check the source to verify that this is the current version before use.



INCHES	MM
.010	0.25
.015	0.38
.250	6.35
.400	10.16
.500	12.70

NOTES:
1. For additional die dimensions see tables I and II.
2. All dies shall have a full length keyway. When the "C" dimension is less than .350 (8.89 mm), the .500 (12.70 mm) maximum cutout shall not extend across the face of the die. Configuration of the cutout is optional.
3. When both A and B hex's require a counterbore, the .250 (6.35 mm) diameter maximum shall apply to the hex with the smallest "D" dimension. The counterbore for the other hex can be larger and shall be shaped so no sharp edges are formed at the intersection of the counterbores.
4. See figures 6, 7 and 8, and tables IV, V, VI and VII for application data.
5. Hex die closure details for coaxial, shielded contacts: Shielded terminating ferrules and coaxial radio frequency, class 2 crimp type series connectors.
6. Dimensions are in inches.
7. Metric equivalents are given for general information only.

FIGURE 3. Typical die configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:46Z
Check the source to verify that this is the current version before use.

TABLE I. Dies for coaxial, shielded contacts.

Part number	Die dimensions						In-service inspection gauge		Contacts accommodated (ref)	Cables accommodated (ref)
	A ±.003(.08) ref section	B ±.003(.08) ref section	C ±.010 (.25)	D ±.010 (.25)	E ±.003 (.08)	F ±.003 (.08)	Die closure A	Die closure B		
M22520/10-02	.091 A"-A" (2.31)	.068 B'-B' (1.73)	.337 (8.56)	.090 (2.29) .270 (6.86)	.125 (3.19)	.093 (2.36)	M22520/5 -030	M22520/6 -029	M39029/6-01 M39029/13-01 M39029/14-01	MIL-C-17/93 Grumman GC 19K-1-22S Revere WH-95523
M22520/10-04	.138 A"-A" (3.51)	.113 B'-B' (3.00)	.337 (8.56)	.090 (2.29) .270 (6.86)	.125 (3.13)	.155 (3.96)	M22520/5 -028	M22520/5 -002	M39029/5-02, -03, -04, -05, -06 M39029/13-02, -03, -04, -05, -06 M39029/14-02, -03, -04, -05, -06	MIL-C-17/94, MIL-C-17/95, MIL-C-17/113 Microdot 293-3922 Microdot 250-4070 Raychem 55300S117 Raychem 5022E5111
M22520/10-05	.150 A-A (4.05)	None	.400 (10.16)	---	.050 (1.27)	---	M22520/6 -005	---	0W089559-1 0W089559-2	MIL-C-17/95
M22520/10-03	.123 A'-A' (3.25)	None	.337 (8.56)	.091 (2.39)	.050 (1.27)	---	M22520/5 -003	---	M39029/7-001 M39029/8-001	MIL-C-17/93 30 AWG 29 AWG
									M39029/7-002 M39029/8-002	MIL-C-17/113 MIL-C-17/119 25 AWG
									M39029/7-003 M39029/8-003	24 AWG
M22520/10-10	.128 A-A (3.25)	None	.375 (9.52)	---	.050 (1.27)	---	M22520/6 -003	---	M39029/50-340 M39029/51-341	MIL-C-17/94

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:46Z
Check the source to verify that this is the current version before use.

TABLE II. Dies for shielded terminating ferrules and for coaxial radio frequency, class 2 crimp type series connectors.

Part no. M22520/10	Die dimensions						In-service inspection gauge	
	A ±.003(.08) ref section	B ±.003(.08) ref section	C ±.010 (.25)	D ±.010 (.25)	E ±.003 (.08)	F ±.003 (.08)	Die closure A	Die closure B
-03	.255 A-A (6.48)	None	.325 (8.26)	---	.050 (1.27)	---	M22520/6 -011	---
-05	.128 A-A (3.25)	.105 B-B (2.68)	.325 (8.26)	---	.102 (2.59)	.108 (2.74)	M22520/6 -003	M22520/6 -001
-07	.213 A-A (5.41)	.178 B-B (4.52)	.325 (8.26)	---	.107 (2.72)	.153 (3.89)	M22520/6 -008	M22520/6 -006
-09	.118 A-A (3.00)	None	.375 (9.53)	---	.030 (0.76)	---	M22520/6 -002	---
-11	.105 A-A (2.67)	None	.400 (10.16)	---	.050 (1.27)	---	M22520/6 -001	---
-13	.128 A-A (3.25)	None	.400 (10.16)	---	.050 (1.27)	---	M22520/6 -003	---
-15	.151 A-A (3.84)	None	.400 (10.16)	---	.050 (1.27)	---	M22520/6 -004	---
-17	.178 A-A (4.52)	None	.400 (10.16)	---	.050 (1.27)	---	M22520/6 -006	---
-19	.197 A-A (5.00)	None	.400 (10.16)	---	.050 (1.27)	---	M22520/6 -007	---
-21	.213 A-A (5.41)	None	.400 (10.16)	---	.050 (1.27)	---	M22520/6 -008	---
-23	.218 A-A (5.54)	None	.400 (10.16)	---	.050 (1.27)	---	M22520/6 -009	---
-25	.231 A-A (5.87)	None	.400 (10.16)	---	.050 (1.27)	---	M22520/6 -010	---
-27	.255 A-A (6.48)	None	.400 (10.16)	---	.050 (1.27)	---	M22520/6 -011	---

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:46Z
Check the source to verify that this is the current version before use.

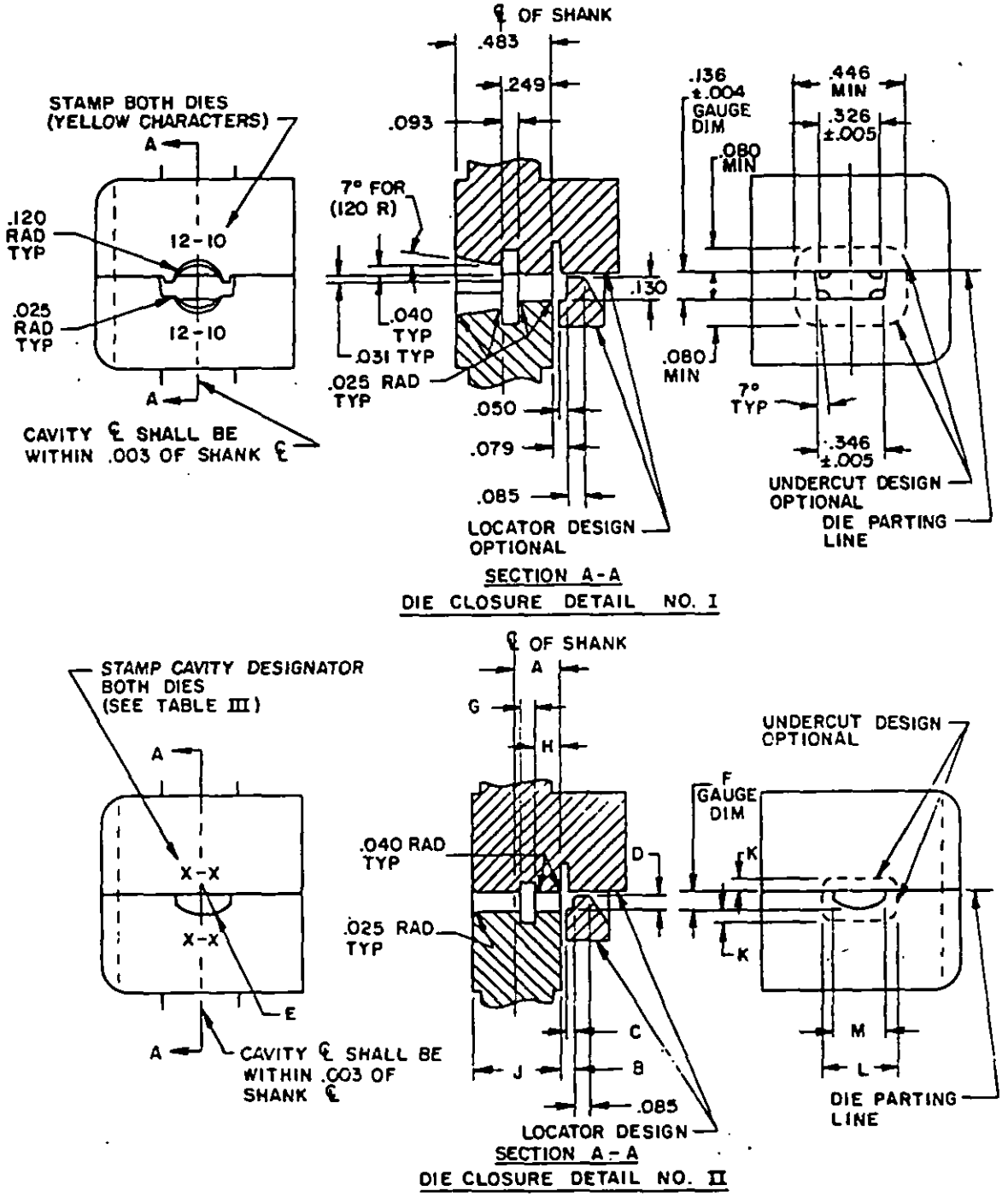


FIGURE 4. Lug terminal and splice die.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:46Z
Check the source to verify that this is the current version before use.

INCHES	MM
.003	0.08
.004	0.10
.005	0.13
.025	0.64
.031	0.79
.040	1.02
.050	1.27
.079	2.01
.080	2.03
.085	2.16
.093	2.36
.120	3.05
.130	3.30
.136	3.45
.249	6.32
.326	8.28
.346	8.79
.446	11.33
.483	12.27

NOTES:
1. Locator is to be attached to upper die. It shall provide a positive stop for locating the item to be crimped and retain it in place without being held by the operator. During the crimping operation the locator shall be free to move as necessary to prevent the crimped item from bending. See Type 2 tools paragraph of MIL-C-22520.
2. *For additional die dimensions see table III.*
3. Die closure details for MS25036, MS17143, M7928 insulated lug terminals and splices, MS25274 and caps. MS27429 insulated disconnect splices and M81824/2 environmental splices.
4. Dimensions are in inches.
5. Metric equivalents are given for general information only.

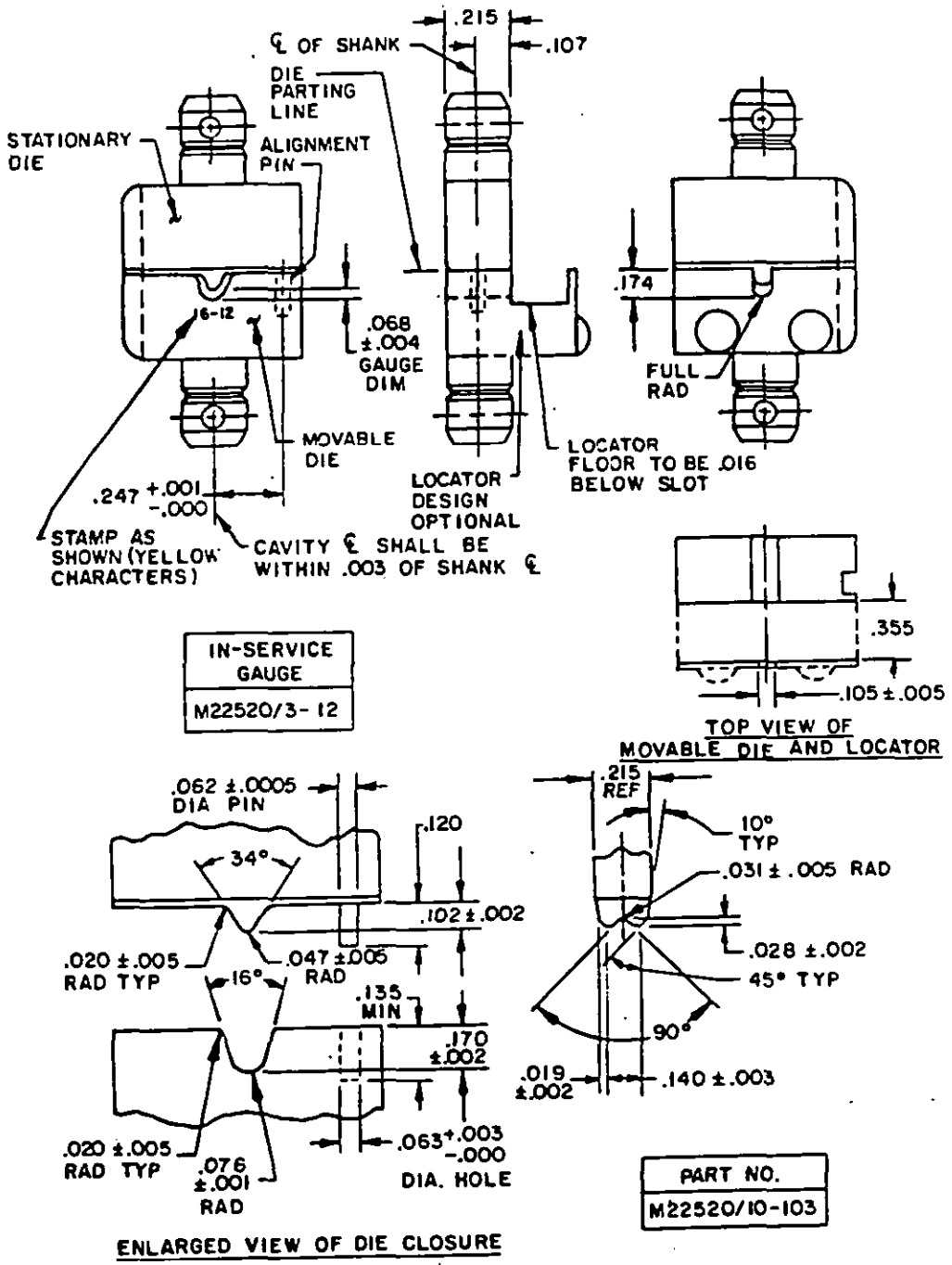
FIGURE 4. Lug terminal and splice die - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:46Z
Check the source to verify that this is the currentversion before use.

TABLE III. Die dimensions for lug terminals and splices.

Part no. M22520/10-	Die size	Detail	A	B	C	D	E (Rad)	F ±.004 (.10)	G	H	I	K min	L min	M ±.005 (.13)	Cavity designator Mark	Cavity designator Color	In-service gauge
100	12-10	I	---	---	---	---	---	---	---	---	---	---	---	---	---	---	M22520/3-9
101	26-14	II	.218 (5.54)	.062 (1.57)	.030 (.76)	.078 (1.98)	.220 (5.59)	.100 (2.54)	.093 (2.36)	.125 (3.18)	.350 (8.89)	.050 (1.27)	.376 (9.55)	.276 (7.01)	26-14	White	M22520/3-10
102	26-20	II	.218 (5.54)	.042 (1.07)	.030 (.76)	.078 (1.98)	.164 (4.17)	.070 (1.78)	.078 (1.98)	.090 (2.29)	.320 (8.13)	.050 (1.27)	.300 (7.62)	.200 (5.08)	26-20 Splice	White	M22520/3-11

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:46Z
Check the source to verify that this is the current version before use.

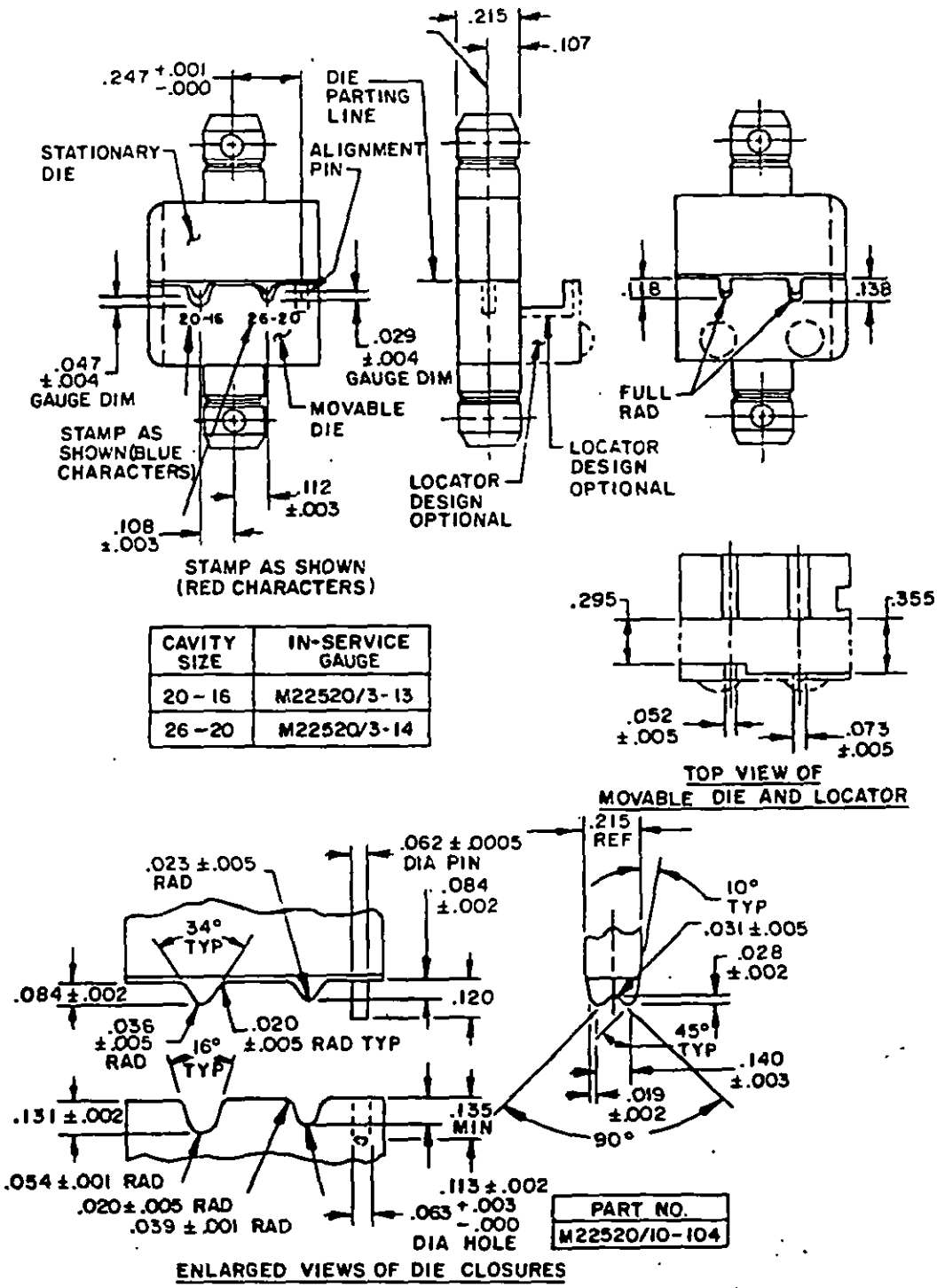


INCHES	MM
.000	0.00
.0005	0.013
.001	0.03
.002	0.05
.003	0.08
.004	0.10
.005	0.13
.016	0.41
.019	0.48
.020	0.51
.028	0.71
.031	0.79
.047	1.19
.062	1.57
.063	1.60
.068	1.73
.076	1.93
.102	2.59
.105	2.67
.107	2.72
.120	3.05
.135	3.43
.140	3.56
.170	4.32
.174	4.42
.215	5.46
.247	6.27
.355	9.02

NOTES:
1. Die faces must touch at completion of crimping cycle.
2. Die closure detail for M81824/1-3 splice.
3. Dimensions are in inches.
4. Metric equivalents are given for general information only.

FIGURE 5. Splice die M22520/10-103.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:46Z
Check the source to verify that this is the current version before use.



CAVITY SIZE	IN-SERVICE GAUGE
20 – 16	M22520/3-13
26 – 20	M22520/3-14

INCHES	MM
.000	0.00
.0005	0.013
.001	0.03
.002	0.05
.003	0.08
.004	0.10
.005	0.13
.019	0.48
.020	0.51
.023	0.58
.028	0.71
.029	0.74
.031	0.79
.036	0.91
.039	0.99
.047	1.19
.052	1.32
.054	1.37
.062	1.57
.063	1.60
.073	1.85
.084	2.13
.107	2.72
.108	2.74
.112	2.84
.113	2.87
.118	3.00
.120	3.05
.131	3.33
.135	3.43
.138	3.51
.140	3.56
.215	5.46
.247	6.27
.295	7.49
.355	9.02

NOTES:
1. Die faces must touch at completion of crimping cycle.
2. Die closure details for M81824/1-1 and M81824/1-2 splices.
3. Dimensions are in inches.
4. Metric equivalents are given for general information only.

FIGURE 6. Splice die M22520/10-104.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:46Z
Check the source to verify that this is the current version before use.

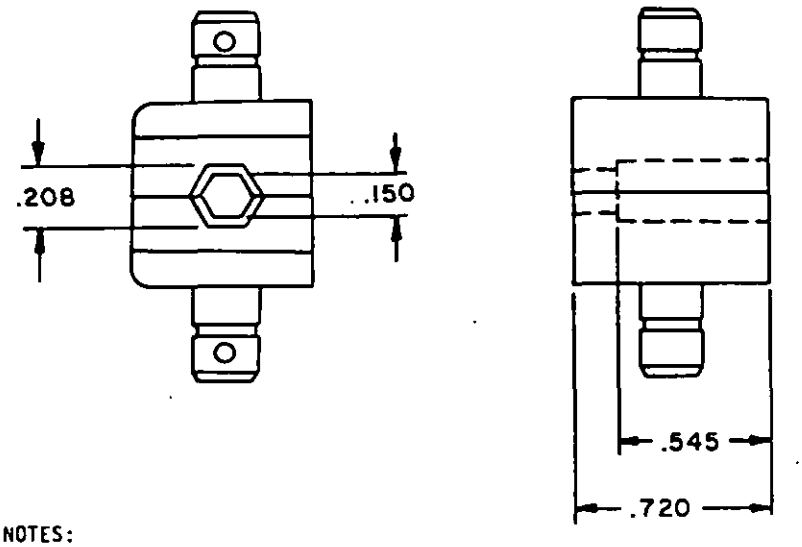
TABLE IV. Application data for insulated terminal lug, splice and end cap dies.

MS standard or spec. sheet	Type of connection	Size	Use M22520/10-01 basic tool with die part no. M22520/10-
MS17143	Terminal	22-18	101
		16-14	101
		12-10	100
MS25036 MIL-T-7928/4	Terminal	26-24	101
		22-18	101
		16-14	101
		12-10	100
MS25274	End cap	26-24	101
		22-18	101
		16-14	101
		12-10	100
MS27429	Splice	22-18	101
		16-14	101
		12-10	100
MIL-T-7928/1	Terminal	26 ↕ 14	101
		12	100
		10	100
MIL-T-7928/2	Terminal	22 ↕ 14	101
		12	100
		10	100
MIL-T-7928/5	Splice	26-24	102
		24-20	102
		22-18	101
		16-14	101
		12-10	100
MIL-S-81824/1	Splice	26-20	104
		20-16	104
		16-12	103
MIL-S-81824/2	Splice	26-20	102
		20-16	101
		16-12	101

TABLE V. Die part number and application data used in fiber optics.

Part number	Connectors accommodated	Cable size	In-service inspection gauge
M22520/10-201	M83522/1 01 and 02	3.0 mm	M22520/6-_ _ _
M22520/10-202	M83522/2 01 and 02	4.5 mm	N22520/6-_ _ _

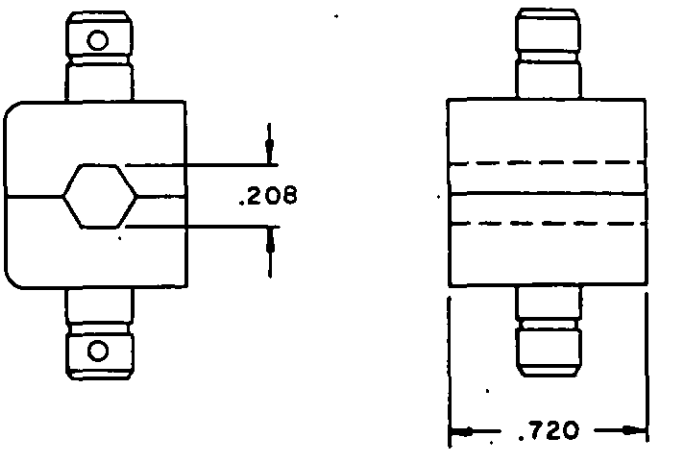
Source: https://assist.dla.mil -- Downloaded: 202610-19T12:46Z
Check the source to verify that this is the current version before use.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Die faces must touch at end of crimping cycle.

FIGURE 7. Fiber optic step die M22520/10-201.

INCHES	MM
.150	3.81
.208	5.28
.545	13.84
.720	18.29



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Die faces must touch at end of crimping cycle.

FIGURE 8. Fiber optic die M22520/10-202.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:46Z
Check the source to verify that this is the current version before use.

TABLE VI. Application data for shield terminating ferrule dies.

Inner conductor insulation dia. range		Inner ferrule part no. MS21981	Outer conductor shield O. D.		To crimp uninsulated outer ferrule part no. MS21980 1/	Use	To crimp insulated outer ferrule part no. MS18121 1/	Use
						Die part no.		Die part no.
Max	Min		Max	Min		M22520/10		M22520/10
.038	.018	-048	.048	.028	-101	-11	-101	-13
.050	.030	-058	.068	.050	-128	-13	-128	-15
.055	.035	-063	.075	.057	-128	-13	-129	-15
.063	.043	-071	.078	.060	-129	-13	-128	-15
.063	.043	-071	.096	.072	-140	-15	-149	-17
.072	.052	-080	.092	.074	-149	-15	-149	-17
.082	.062	-090	.097	.079	-149	-15	-149	-17
.088	.068	-096	.105	.089	-155	-06	-156	-19
.093	.073	-101	.124	.106	-175	-17	-175	-21
.102	.082	-109	.130	.119	-175	-17	-175	-21
.102	.082	-109	.137	.117	-187	-19	-187	-25
.116	.096	-124	.138	.118	-187	-19	-187	-25
.116	.095	-124	.146	.128	-194	-19	-194	-25
.116	.096	-124	.151	.139	-199	-19	-199	-25
.120	.100	-128	.153	.133	-205	-21	-205	-27
.120	.100	-128	.155	.147	-219	-21	-219	-27
.120	.106	-134	.143	.128	-205	-21	-205	-27
.120	.106	-134	.163	.149	-225	-23	-225	-27
.126	.106	-134	.191	.162	-232	-25		
.141	.121	-149	.171	.153	-232	-25		
.148	.128	-155	.192	.174	-261	-27		
.157	.137	-165	.195	.177	-261	-27		
.167	.147	-175	.199	.165	-261	-27		

1/ Not preferred for Army use.

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:46Z
Check the source to verify that this is the current version before use.

TABLE VII. Crimp tool application data for coaxial radio frequency, class 2 crimp type series connector.

MIL-C-39012 connector part no. M39012/	To crimp 1/ inner contact part no. M39012/	Use 2/ turret part no. M22520/1	To crimp outer ferrule part no. M39012/	Use 3/ die part no. M22520/5	Cables accommodated (ref.) MIL-C-17/
1-0503	1-11	-13	1-52	-21	60,128
2-0503	2-11	-13	2-52	-21	60,128
2-0513	2-11	-13	2-52	-21	60,128
3-0503	3-11	-14	3-52	-21	60,128
5-0503	5-11	-13	5-52	-21	60,128
16-0501	16-12	-12	16-50	-17	54
16-0502	16-11	-12	16-50	-17	95
16-0503	16-10	-12	16-51	-21	60,128
17-0501	17-12	-12	17-50	-17	54
17-0502	17-11	-12	17-50	-17	95
17-0503	17-10	-12	17-51	-21	60,128
18-0501	18-12	-12	18-50	-17	54
18-0502	18-11	-12	18-50	-17	95
18-0503	18-10	-12	18-51	-21	60,128
19-0501	19-12	-12	19-50	-17	54
19-0502	19-11	-12	19-50	-17	95
19-0503	19-10	-12	19-51	-21	60,128
20-0501	20-12	-12	20-50	-17	54
20-0502	20-11	-12	20-50	-17	95
20-0503	20-10	-12	20-51	-21	60,128
26-0501	26-12	-12	26-50	-17	54
26-0502	26-11	-12	26-50	-17	95
26-0503	26-10	-12	26-51	-21	60,128
27-0501	27-12	-12	27-50	-17	54
27-0502	27-11	-12	27-50	-17	95
27-0503	27-10	-12	27-51	-21	60,128
28-0501	28-12	-12	28-50	-17	54
28-0502	28-11	-12	28-50	-17	95
28-0503	28-10	-12	28-51	-21	60,128
29-0501	29-12	-12	29-50	-17	54
29-0502	29-11	-12	29-50	-17	95
29-0503	29-10	-12	29-51	-21	60,128
30-0501	30-12	-12	30-50	-17	54
30-0502	30-11	-12	30-50	-17	95
30-0503	30-10	-12	30-51	-21	60,128
35-0503	35-11	-14	35-51	-21	60,128
36-0503	36-11	-14	36-51	-21	60,128
38-0503	38-11	-14	38-51	-21	60,128
39-0503	39-11	-13	39-51	-21	60,128
40-0503	40-11	-14	40-51	-21	60,128
55-4502	55-10	-15	55-50	-21	60,128
56-4502	56-10	-15	56-50	-21	60,128
57-4502	57-10	-15	57-50	-21	60,128
58-4502	58-10	-15	58-50	-21	60,128
59-4502	59-10	-15	59-50	-21	60,128

1/ Not preferred for Army use.
2/ Use indent crimping tool M22520/1-01 with turret.
3/ Use basic tool M22520/10-01.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T12:46Z
Check the source to verify that this is the current version before use.

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, AV, MI
Navy - EC, OS, SH
Air Force - 12, 99

User activities:
Army - AT, GL, ME
Navy -
Air Force -

Preparing activity:
Navy - AS

Agent:
DLA - ES

(Project 5120-B051)

Source: https://assist.dla.mil -- Downloaded: 202610-19T12:46Z
Check the source to verify that this is the current version before use.